Quoc Huy Pham

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Portfolio

Education:

Bachelor of Engineering (Honours) Macquarie University, Sydney, Australia
Specialisation in Software Engineering February 2022 – February 2026
Relevant coursework: System Design and Development, Data Structure and Algorithm, Scientific Computing, Embedded System.

Diploma of Information Technology Macquarie University, Sydney, Australia
Foundation in programming fundamentals and IT concepts February 2021 – February 2022

Experiences:

VietJapan Digital Ho Chi Minh City, Vietnam
Software Engineer Intern Jun 2025 – Aug 2025

- Gained hands-on experience in full-stack development working on confidential client projects
- Built and maintained an NDA backend service using Node.js and Express.js framework
- Collaborated with cross-functional teams in an Agile development environment
- Participated in code reviews and implemented feedback to improve code quality
- Documented technical specifications and maintained project documentation

Solutions 8 Sydney, Australia
General IT Support and Web Designer Jun 2023 – Jun 2025

- Provided technical support for individuals and small businesses.
- Diagnosed and resolved hardware, software, and networking issues for a diverse client base
- Provided technical consultations and recommendations for small business IT solutions
- Designing and building business websites through WordPress

Technical Skills:

- **Frontend Development:** JavaScript, TypeScript, HTML5, CSS3, React, Next.js, Tailwind CSS, Shadcn/ui, Zustand
- **Backend Development:** Node.js, Express.js, Java, MATLAB, Python
- **Embedded System:** C, Arduino
- **Databases & Caching:** MySQL, Redis
- **Data Science & Machine Learning:** Pandas, PyTorch, Scikit-learn, Label Studio
- **Tools:** Git, Docker, Cursor, Claude Code, Figma, Microsoft Office Suite
- **Operating Systems:** Linux, Windows, macOS

Academic Projects:

NLP-Based Privacy Requirement Classifier Thesis – Academic Year 2025

Objective: Automate detection of GDPR-related privacy requirements in software specifications using fine-tuned transformer models, targeting resource-constrained small to medium businesses.

Tools: Python, PyTorch, Hugging Face Transformers, Label Studio, Scikit-learn

Responsibilities:

- Fine-tuned and benchmarked three transformer architectures (DistilBERT, MiniLM, XLNet) or binary classification on consumer-grade hardware (6GB VRAM)
- Curated and labelled 1,476 software requirements using a GDPR-inspired Privacy-by-Design framework with 7 labelling criteria
- Addressed 70% class imbalance through CLAIRE-based data augmentation, improving minority class representation from 12% to 29%
- Designed end-to-end ML pipeline: data cleaning, stratified splitting, tokenization, training with early stopping, and evaluation

Outcome: Best model (XLNet) achieved 0.955 F1-score and 98.5% recall, demonstrating production-ready privacy compliance automation without cloud infrastructure

VivaMQ - AI-Powered Viva Question Generator Academic Year 2024

Objective: Develop a web application to automatically generate viva examination questions and marking guidelines using AI

Tools: Next.js, TypeScript, Tailwind CSS, Zustand, Shadcn/ui, GPT API

Responsibility:

- Served as Front-End Technical Lead, coordinating development efforts across the development teams
- Architected the application structure using Next.js with TypeScript for type safety
- Implemented state management solution using Zustand for efficient data flow
- Designed responsive UI components using Tailwind CSS and Shadcn/ui
- Delivered a functional application that helps tutors efficiently create assessment materials

Outcome: Delivered a functional application that helps tutors efficiently create assessment materials

Marble Machine Academic Year 2023

Objective: Develop embedded software for a Rube Goldberg marble machine with integrated electronic components

Tools: Arduino, C programming, Embedded systems

- Collaborated as part of the communication team to develop and manage embedded software
- Programmed microcontrollers to control various electronic components within the marble machine
- Ensured proper system integration and communication between different machine modules
- Tested and debugged embedded code to ensure reliable operation

Outcome: Successfully delivered a fully functional marble machine with seamless electronic integration

Weather Data Analysis in Scientific Computing Academic Year 2022

Objective: Create a command-line interface application for weather data processing and statistical analysis Tools: MATLAB, Excel

Responsibilities:

- Developed algorithms for data cleaning and preprocessing of weather datasets
- Implemented statistical analysis functions for temperature, precipitation, and weather patterns
- Created data visualization tools for presenting analysis results
- Designed efficient data manipulation techniques using MATLAB's built-in functions

Outcome: Gained foundational skills in data processing and scientific computing